Kinematics and integration

1

a $x(t) = 6t^2$

b 3 s

2

a $s(t) = 3t^2 + 10t + 8$

b 133 m

3

a $s(t) = 4t^3 + 15t^2 + 9t - 6$

b 914 m

4

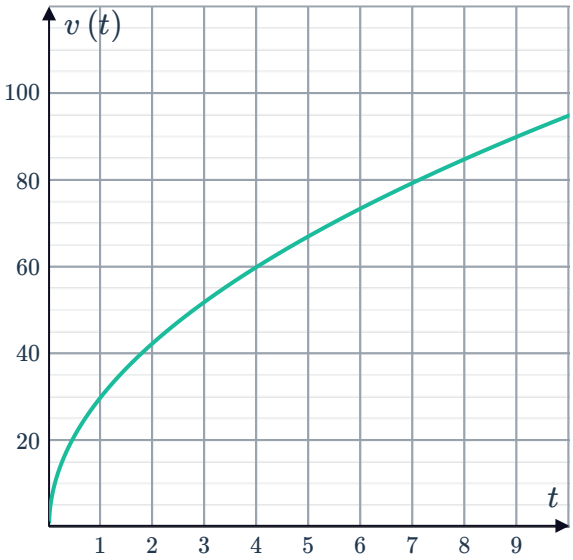
a $x(t) = 8\sqrt{t^3} + 5$

b 221 m to the right of the origin.

5

a 60 m/s

b



c 160 units^2

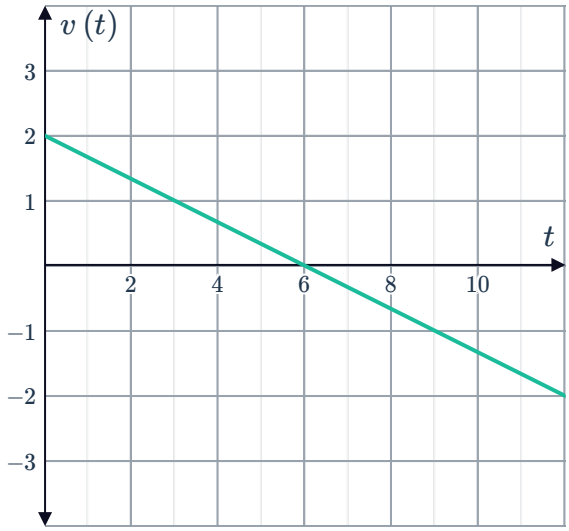
d The distance travelled by the particle in the first 4 seconds. As the function is positive over this interval the area also represents the displacement of the object from its initial position after 4 seconds.

6

a -1 m/s

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b



c 7.5 units^2

d The distance travelled by the particle in the first 9 seconds.

7

a $15\,\text{m/s}$

b $x\,(t) = 3t^2 + 15t + 3$

c 111 m

d 66 m

8

a 1 s

b $x\,(t) = 2t^2 - 4t + 3$

c

i 3 **ii** 1 **iii** 3

9

a $3\,\text{m/s}$

b $x\,(t) = 2t^3 - 7t^2 + 3t + 2$

c 30 m

d 34 m

10

a $x\,(t) = 4t^3 - 24t^2 + 5$

b $t = 0, 4$ s

c -123 m

11

a $-2\,\text{m/s}^2$

b $x = \dfrac{6t^{\frac{5}{2}}}{5} - t^2$

c $t = \dfrac{4}{9}$ s

d 2.87 m

12

a $x\,(t) = t^4 - 12t^3 + 36t^2 + 5$

b A turning point. At $t = 3$ the pen changes direction and starts moving to the left.

c 1.96 mg

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13

a $5\left(4+e^{-5}\right)$ m

b 4.007 m

c 4.007 m

d Yes, because the velocity is always positive for $t \geq 0$.

14

a $v(t)=t^2-15t+56$

b $t=8,7$ s

15

a $v(t)=3t^2-27t+60$

b $t=4,5$ s

16

a $v(t)=2t-5$

b $x(t)=t^2-5t+8$

c 22 m to the right of the origin.

d $t=4$ s

17

a $t=2$ s

b 50 m

c $t=2+\sqrt{10}$ s

d $-10\sqrt{10}$ m/s

18

a $t=\frac{75}{49}$ s

b 13.5m

c $t=3.2$s

d -16.4 m/s

19

a $v(t)=-\frac{1}{4}e^{4t}+\frac{1}{4}$

b $-30\,322\,823$ cm

20

a 2.98 m/s

b 11.02 m

21

a 278 m/s

b $36\,203$ m

Trigonometric applications

22

a -18 m/s^2

b 3 m

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23

a $-9\sqrt{3}\ \text{m/s}^2$

b $3\ \text{m}$

24

a $\dfrac{18}{\pi}\ \text{m}$

b $\dfrac{30}{\pi}\ \text{m}$

25

a $\dfrac{12\sqrt{2}}{\pi}\ \text{m}$

b $\dfrac{96}{\pi}\ \text{m}$

26

a $v(t) = -16\cos 4t + 18$

b $26\ \text{m/s}$

c $x(t) = -4\sin 4t + 18t + 3$

d $\left(2\sqrt{3} + 6\pi + 3\right)\ \text{m}$

27

a $v(t) = 10\sin 5t + 6$

b $11\ \text{m/s}$

c $x(t) = -2\cos 5t + 6t + 5$

d $\left(\sqrt{3} + \pi + 5\right)\ \text{m}$

28

a $0\ \text{m/s}$

b $\dfrac{45}{\pi}\ \text{m/s}$

29

a $0\ \text{m/s}$

b $\dfrac{32}{\pi}\ \text{m/s}$

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